Intake valve timing:

The inlet valve will be open 10° to 30° before TDC (i.e during exhaust stroke), remain open during suction stroke and closes at 30° to 40° after BDC (i.e. during compression stroke). This gives the inlet valve a total opening of 220° of crankshaft rotation. This will ensure the full induction of fresh charge in cylinder during suction stroke.

After the inlet valve closed in the compression stroke, the air fuel mixture is compressed by the piston. This will increase the pressure and temperature in the cylinder.

With both intake and exhaust valve are in closed condition, the spark will be introduced by the ignition system through spark plug at 20° to 40° before TDC. These will initialise the combustion and the air fuel mixture will be burned in the combustion chamber. Thus the chemical energy of the fuel is converted into heat energy.

Exhaust valve timing:

The exhaust valve opens at 30° to 60° before the completion of power stroke. Due to this, the gases have an outlet for expansion, which removes the greater part of the burnt gases. The valves remain open during open during the exhaust stroke and it is closed at 20° after TDC, during the intake stroke of the next cycle.

Scavenaging

The inlet valve will be open 10° to 30° before TDC and at the same time the exhaust valve is already in the open condition. This will make the fresh charge to push out the burnt out gases from the cylinder. Scavenging will be made till 20°After TDC.

Valve over Lap

The portion of the operating cycle in which, when the piston is passing TDC (top dead centre) on the exhaust stroke, both the intake and exhaust valves are open. It is usually expressed in degrees of crankshaft rotation. Valve overlap is necessary for the efficient flow of gases in and out of the combustion chamber.

5.5 TWO STROKE PETROL ENGINE

The two stroke cycle engine completes one power stroke in one revolution of a crankshaft.

Construction

Inside the engine, the small end of the connecting rod is connected to the piston with the help of piston pin. The big end of the connecting rod is connected to the crankpin of the crankshaft. When the crankshaft rotates, the piston will reciprocate on the cylinder and vice versa. There is no inlet and exhaust valves as in case of four-stroke engine but consists of the inlet port (IP), an exhaust port (EP) and transfer port (TP). The fresh charge enters into the crankcase through inlet port. Transfer port is used to transfer the compressed charge from crankcase to the cylinder. Exhaust port is used to transfer the burnt gases out of the engine. The movement of the piston will open and closes the ports. A spark plug will be placed in the engine head.

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